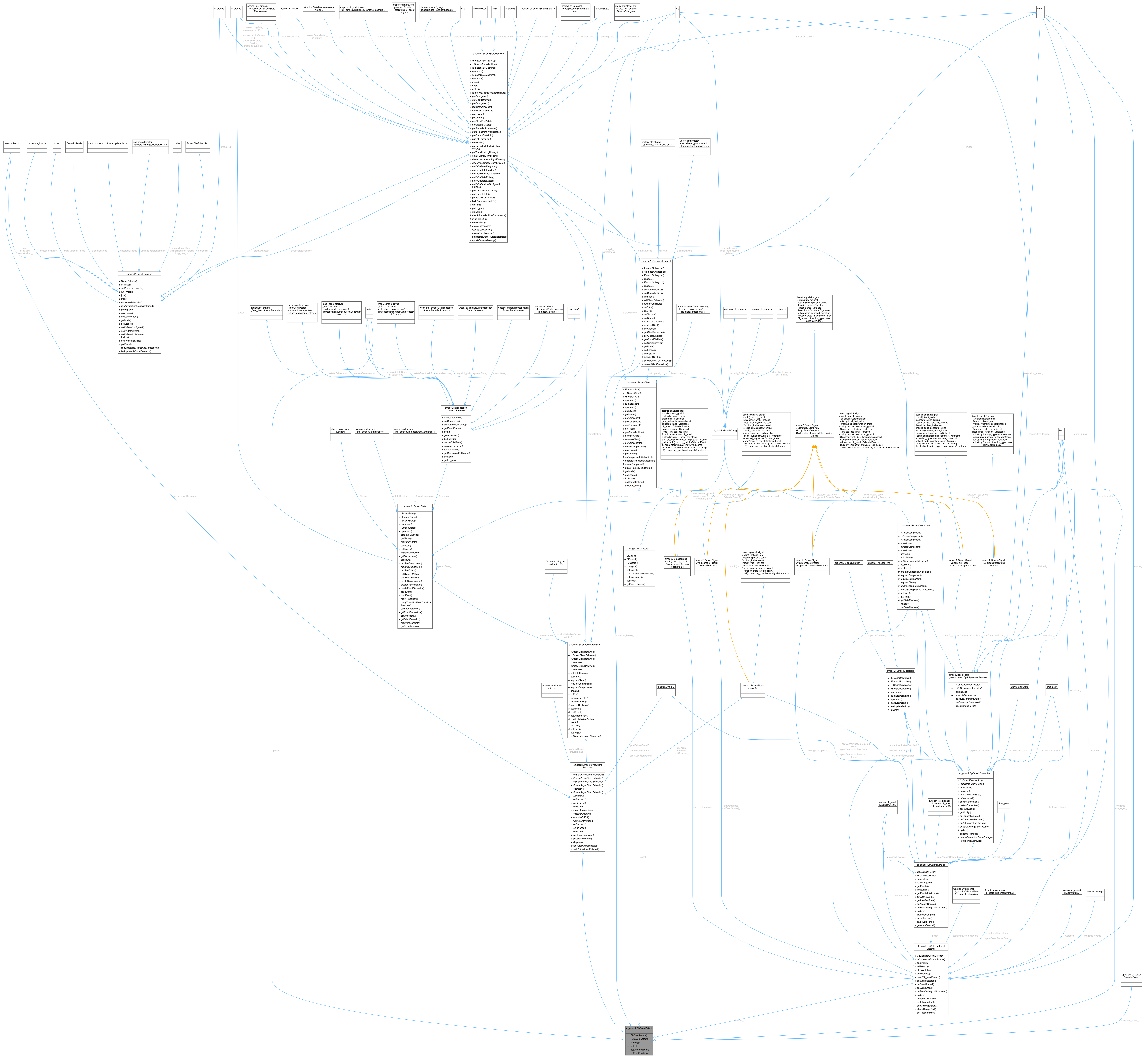
SharedPtr
SharedPtr
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
recursive_mutex
atomic< StateMachineInternalAction >
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function< std::string()>, boost::any > >
deque< smacc2_msgs::msg::SmaccTransitionLogEntry >
size_t
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2::introspection::SmaccStateInfo >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
int
mutex
#eventsLogPub_
#stateMachinePub_
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
#nh_
#stateMachineInfo_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-transitionLogHistorySize_
-runMode_
-stateSeqCounter_
#timer_
#currentState_
#currentStateInfo_
#status_msg_
#orthogonals_
-reactorWalkDepth_
-transitionLogMutex_
-mutex_
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ ISmaccStateMachine()
+ operator=()
+ ISmaccStateMachine()
+ operator=()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ onUnhandledStInitializationFailure()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
atomic< bool >
processor_handle
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector< smacc2::ISmaccUpdatable * > >
double
SmaccFifoScheduler *
-statusPub_
-end_
-initialized_
-rosInitialized_
-processorHandle_
-signalDetectorThread_
-executionModel_
-updatableClients_
-updatableStateElements_
-kDefaultLoopRateHz
-kInitializationPollRateHz
-loop_rate_hz
-scheduler_
-signalDetector_
-smaccStateMachine_
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyStateInitializationFailed()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
vector< std::shared_ptr< smacc2::ISmaccClient > >
vector< std::vector< std::shared_ptr< smacc2::ISmaccClientBehavior > > >
+depth_
+stateIndex_
-stateMachine_
#clients_
-clientBehaviors_
+agenda_days
+max_consecutive_failures
smacc2::ISmaccOrthogonal
+ ISmaccOrthogonal()
+ ~ISmaccOrthogonal()
+ ISmaccOrthogonal()
+ operator=()
+ ISmaccOrthogonal()
+ operator=()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
std::enable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorInfoEntry > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
string
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
weak_ptr< smacc2::introspection::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection::SmaccStateInfo >
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
optional< std::string >
vector< std::string >
seconds
#node_
+staticBehaviorInfo
+eventGeneratorsInfo
+demangledStateName
+fullStateName
+stateReactorsInfo
+stateMachine_
+gcalcli_path
+parentState_
+transitions_
+children_
+tid_
-orthogonal_
#components_
+config_folder
+calendars
+heartbeat_interval
+poll_interval
boost::signals2::signal< Signature, optional_last_value< typename boost::function_traits< Signature >::result_type >, int, std::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost::signals2::mutex >
smacc2::introspection::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ ISmaccClient()
+ operator=()
+ ISmaccClient()
+ operator=()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
boost::signals2::signal< void(const cl_gcalcli::CalendarEvent &, const std::string &), optional_last_value< typename boost::function_traits< void(const cl_gcalcli::CalendarEvent &, const std::string &)>::result_type >, int, std::less< int >, function< void(const cl_gcalcli::CalendarEvent &, const std::string &)>, typename extended_signature< function_traits< void(const cl_gcalcli::CalendarEvent &, const std::string &)>::arity, void(const cl_gcalcli::CalendarEvent &, const std::string &)>::function_type, boost::signals2::mutex >
cl_gcalcli::GcalcliConfig
boost::signals2::signal< void(const cl_gcalcli::CalendarEvent &), optional_last_value< typename boost::function_traits< void(const cl_gcalcli::CalendarEvent &)>::result_type >, int, std::less< int >, function< void(const cl_gcalcli::CalendarEvent &)>, typename extended_signature< function_traits< void(const cl_gcalcli::CalendarEvent &)>::arity, void(const cl_gcalcli::CalendarEvent &)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal< Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal< void(const std::vector< cl_gcalcli::CalendarEvent > &), optional_last_value < typename boost::function_traits < void(const std::vector< cl_gcalcli::CalendarEvent > &)>::result_type >, int, std::less< int >, function < void(const std::vector< cl_gcalcli::CalendarEvent > &)>, typename extended_signature< function_traits< void(const std::vector< cl_gcalcli::CalendarEvent > &)>::arity, void(const std::vector< cl_gcalcli::CalendarEvent > &)>::function_type, boost::signals2::mutex >
boost::signals2::signal< void(int exit_code, const std::string &output), optional_last_value< typename boost::function_traits< void (int exit_code, const std::string &output)>::result_type >, int, std::less< int >, function< void(int exit_code, const std::string &output)>, typename extended_signature< function_traits< void (int exit_code, const std::string &output)>::arity, void(int exit_code, const std::string &output)>::function_type, boost::signals2::mutex >
boost::signals2::signal< void(const std::string &error), optional_last_value< typename boost::function_traits< void(const std::string &error)>::result_type >, int, std::less< int >, function< void(const std::string &error)>, typename extended_signature< function_traits< void(const std::string &error)>::arity, void(const std::string &error)>::function_type, boost::signals2::mutex >
#stateMachine_
-execution_mutex_
-consecutive_failures_
-state_mutex_
bool
shared_ptr< rclcpp::Logger >
vector< std::shared_ptr< smacc2::StateReactor > >
vector< std::shared_ptr< smacc2::SmaccEventGenerator > >
#logger_
#stateReactors_
#eventGenerators_
#stateInfo_
-currentOrthogonal
-config_
< void(const cl_gcalcli::CalendarEvent &, const std::string &)>
< void(const cl_gcalcli::CalendarEvent &)>
#initializationFailed_
#owner_
< void(const std::vector< cl_gcalcli::CalendarEvent > &)>
< void(int exit_code, const std::string &output)>
< void(const std::string &error)>
-events_mutex_
-isShutdownRequested_
smacc2::ISmaccState
+ ISmaccState()
+ ~ISmaccState()
+ ISmaccState()
+ operator=()
+ ISmaccState()
+ operator=()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ initializationFailed()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransitionTypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
cl_gcalcli::ClGcalcli
+ ClGcalcli()
+ ClGcalcli()
+ ~ClGcalcli()
+ configure()
+ getConfig()
+ onComponentInitialization()
+ getConnection()
+ getPoller()
+ getEventListener()
smacc2::SmaccSignal< void(const cl_gcalcli::CalendarEvent &, const std::string &)>
smacc2::SmaccSignal< void(const cl_gcalcli::CalendarEvent &)>
boost::signals2::signal< void(), optional_last_value< typename boost::function_traits< void()>::result_type >, int, std::less< int >, function< void()>, typename extended_signature< function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >
< void()>
smacc2::SmaccSignal< void(const std::vector< cl_gcalcli::CalendarEvent > &)>
optional< rclcpp::Duration >
optional< rclcpp::Time >
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ ISmaccComponent()
+ operator=()
+ ISmaccComponent()
+ operator=()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
smacc2::SmaccSignal< void(int exit_code, const std::string &output)>
smacc2::SmaccSignal< void(const std::string &error)>
-initialized_
-mutex_
function< void(const std::string &)>
-currentState
-postInitializationFailureEventFn_
-minutes_before_
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ ISmaccClientBehavior()
+ operator=()
+ ISmaccClientBehavior()
+ operator=()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# postInitializationFailureEvent()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
optional< std::future< int > >
function< void()>
smacc2::SmaccSignal< void()>
-periodDuration_
-lastUpdate_
-config_
+onCommandCompleted_
+onCommandFailed_
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ ISmaccUpdatable()
+ operator=()
+ ISmaccUpdatable()
+ operator=()
+ executeUpdate()
+ setUpdatePeriod()
# update()
smacc2::client_core_components::CpSubprocessExecutor
+ CpSubprocessExecutor()
+ ~CpSubprocessExecutor()
+ onInitialize()
+ executeCommand()
+ executeCommandAsync()
+ onCommandCompleted()
+ onCommandFailed()
ConnectionState
time_point
-onEntryThread_
-onExitThread_
-postFailureEventFn
-postFinishEventFn
-postSuccessEventFn_
-onFailure_
-onFinished_
-onSuccess_
+onAgendaUpdated_
+postAuthenticationRequiredEvent_
+postConnectionLostEvent
+postConnectionRestoredEvent_
+onAuthenticationRequired
+onConnectionLost
+onConnectionRestored_
-subprocess_executor_
-connection_state_
-last_heartbeat_time_
-last_poll_attempt_
-triggered_
-use_regex_
smacc2::SmaccAsyncClientBehavior
+ onStateOrthogonalAllocation()
+ SmaccAsyncClientBehavior()
+ ~SmaccAsyncClientBehavior()
+ SmaccAsyncClientBehavior()
+ operator=()
+ SmaccAsyncClientBehavior()
+ operator=()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()
cl_gcalcli::CpGcalcliConnection
+ CpGcalcliConnection()
+ ~CpGcalcliConnection()
+ onInitialize()
+ configure()
+ getConnectionState()
+ isConnected()
+ checkConnection()
+ restartConnection()
+ executeGcalcli()
+ getConfig()
+ onConnectionLost()
+ onConnectionRestored()
+ onAuthenticationRequired()
+ onStateOrthogonalAllocation()
# update()
- performHeartbeat()
- handleConnectionStateChange()
- isAuthenticationError()
vector< cl_gcalcli::CalendarEvent >
function< void(const std::vector< cl_gcalcli::CalendarEvent > &)>
time_point
+onEventDetected_
+onEventEnded_
+onEventStarted_
-cached_events_
+postAgendaUpdatedEvent_
-connection_
-last_poll_time_
cl_gcalcli::CpCalendarPoller
+ CpCalendarPoller()
+ ~CpCalendarPoller()
+ onInitialize()
+ refreshAgenda()
+ getEvents()
+ findEvents()
+ getEventsInWindow()
+ getActiveEvents()
+ getLastPollTime()
+ onAgendaUpdated()
+ onStateOrthogonalAllocation()
# update()
- parseTsvOutput()
- parseTsvLine()
- parseDateTime()
- generateEventId()
-current_events_
function< void(const cl_gcalcli::CalendarEvent &, const std::string &)>
function< void(const cl_gcalcli::CalendarEvent &)>
vector< cl_gcalcli::EventWatch >
set< std::string >
-poller_
+postEventDetectedEvent_
+postEventEndedEvent
+postEventStartedEvent_
-watches_
-triggered_events_
cl_gcalcli::CpCalendarEventListener
+ CpCalendarEventListener()
+ ~CpCalendarEventListener()
+ onInitialize()
+ addWatch()
+ clearWatches()
+ getWatches()
+ resetTriggeredEvents()
+ onEventDetected()
+ onEventStarted()
+ onEventEnded()
+ onStateOrthogonalAllocation()
# update()
- onAgendaUpdated()
- matchesPattern()
- shouldTriggerStart()
- shouldTriggerEnd()
- getTriggeredKey()
optional< cl_gcalcli::CalendarEvent >
-pattern_
-client_
-listener_
-detected_event_
cl_gcalcli::CbEventDetect
+ CbEventDetect()
+ ~CbEventDetect()
+ onEntry()
+ onExit()
+ getDetectedEvent()
- onEventStarted()